# AYLA DESAI

San Diego, California • 858-250-9779 • ayla0desai@gmail.com • LinkedIn

## SUMMARY

Mechanical Engineering sophomore at Cornell University (4.09 GPA) with hands-on design and manufacturing experience through Cornell Baja Racing. Proficient in SolidWorks, Python, and manufacturing techniques. Seeking a Summer 2027 Mechanical Engineering Internship.

## EDUCATION

**Cornell University,** Ithaca, New York *Sophomore, Graduating May 2029*

- B.S. Mechanical Engineering|GPA: 4.09
- **Relevant Coursework:** Physics Mechanics, Differential Equations, Multivariable Calculus, Computer Science for Developers; *In Progress:* Statics, Thermodynamics, Physics E&M, Linear Algebra

**La Jolla Country Day School,** San Diego, California *Graduated May 2025*

- **Academic Performance:** 4.78 weighted GPA, 3.99 unweighted GPA
- **Academic Honors:** Valedictorian, Cum Laude Society Inductee, Mission Federal Scholar Athlete of the Year

## ENGINEERING & TECHNICAL EXPERIENCE

**Cornell Baja Racing — Brakes Subteam Member** *Fall 2025 – Expected Spring 2029*

- Collaborated with a team of 50+ students to design, manufacture, and race an off-road car against 100+ national and international collegiate teams.
- Leading master cylinder redesign for 2026–2027; this includes diagnosing prior year's brake drag and force loss issues, designing new master cylinders to address these issues, and manufacturing and assembling this season's cylinders.
- Designed and manufactured the firewall, spillguard, and non-structural mounting tabs for the team's off-road vehicle in the 2025-2026 season.
- Modeled each part in SolidWorks and produced manufacturing drawings; hand-cut and fit sheet aluminum components to the frame within a quarter-inch tolerance to meet SAE standards.

**Naval Information and Warfare Center — SEAP Intern** *June 2024 – August 2024*

- Collaborated with a team of 8 interns to evaluate leach rates of Navy anti-fouling coatings, applying analytical chemistry techniques to assess environmental impact.
- Constructed a testing rig based on an established design; operated an ICP mass spectrometer to quantify mercury concentration and evaluated salt-absorption capacity across multiple resin types.

**C. Elegans Worm Pick Design — Personal Project** *July 2026*

- Designed a 3D-printable worm pick for a C. elegans research lab at UCSD to address flimsiness and instability in commercial picks; iterated through 8 prototype versions in Onshape to improve grip and durability.
- Reduced manufacturing cost from $130 (commercial) to $0.50 per unit while enabling mass production, and added an adjustable platinum wire length; a feature unavailable in market alternatives.

## TEACHING & LEADERSHIP EXPERIENCE

**Cornell University — Introduction to Mechanical Engineering TA** *August 2026 – Expected December 2026*

- Held office hours and supported lab sessions for an engineering course of 50-60 students.

**Cornell University — Academic Excellence Workshop Instructor** *January 2026 – May 2026*

- Taught weekly multivariable calculus sessions to groups of 15–20 engineering students, designing original presentations and review materials to promote active learning.
- Completed weekly leadership training to strengthen instructional strategies and improve student engagement.

## ADDITIONAL ACTIVITIES

**Less Than Three Acapella — Member** *September 2025 – April 2026*

**Competitive FTC Robotics — Definitely Human, Team 19743** *Fall 2021 – Spring 2025*

- Achieved peak rank of 6th overall and 2nd in autonomous out of 9000 teams worldwide.

## TECHNICAL PROFICIENCIES

- **CAD and FEA:** SolidWorks, Onshape, Fusion, Ansys | **Programming:** Python, Java | **Foreign Languages:** Spanish
- **Manufacturing/Lab:** Sheet metal fabrication, hand tools, materials testing, ICP-MS, CNC, 3D printing, manual mills, manual lathe, composite layups.